

Full Stack Web Development Lab

Week 9 - Record

Aim

To develop a user interface for a real-world scenario using React (JSX – JavaScript Extension), namely Premium Fix, a home services booking application (Plumber, Home Cleaning, Laundry and Electrician), with a Registration Form with validation.

Description

Premium Fix is a single-page React application built with Vite in which users book home services online. The page is divided into reusable function components: App (navigation, page switching and shared state), ServiceCard (one card per service), RegisterForm (registration with validation) and BookingForm (booking pop-up). The service details (name, icon, price, duration, description) are kept in data.js and displayed as cards using the map() function. The Registration Form collects full name, email, 10-digit mobile number, city, password, confirm password and a terms checkbox; the values are held in state with useState, and on submit a validate() function checks every field (name length, email pattern, mobile pattern, city selected, minimum password length, password match and terms accepted) and shows an error message under each wrong field. After successful registration the navbar greets the user by name. Clicking Book on a service opens a pop-up where the user picks the date, time slot and address; if the user has not registered, the pop-up asks to register first. A confirmed booking is added to the bookings state and listed under My bookings, from where it can be cancelled. Props such as onBook, onDone, onConfirm and onClose are used to pass data and actions between components. Flow: Landing Page → Registration Form → Validate Fields → Select Service → Booking Form → Confirm Booking → My Bookings → Cancel.

Program

package.json

```
{
  "name": "premium-fix",
  "private": true,
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^18.3.1",
    "react-dom": "^18.3.1"
  },
  "devDependencies": {
    "@vitejs/plugin-react": "^4.3.1",
    "vite": "^5.4.0"
  }
}
```

vite.config.js

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react";
export default defineConfig({
  plugins: [react()]
});
```

index.html (HTML entry page)

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Premium Fix - Home Services</title>
    <link href="https://fonts.googleapis.com/css2?family=Bricolage+Grotesque:wght@600;800&family=Manrope:wght@400;600;700&display=swap" rel="stylesheet" />
  </head>
  <body><div id="root"></div><script type="module" src="/src/main.jsx"></script></body>
</html>
```

src/main.jsx (React entry point)

```
import React from "react";
import { createRoot } from "react-dom/client";
import App from "../App.jsx";
import "../styles.css";
createRoot(document.getElementById("root")).render(<App />);
```

src/data.js (services data)

```
export const services = [
  { id: 1, name: "Plumber", icon: "■", price: 299, time: "60 min", desc: "Leaks, taps, blocked drains and pipe fitting." },
  { id: 2, name: "Home Cleaning", icon: "■", price: 799, time: "3 hrs", desc: "Deep clean for kitchen, bathrooms and rooms." },
  { id: 3, name: "Laundry", icon: "■", price: 149, time: "24 hrs", desc: "Wash, dry and fold with doorstep pickup." },
  { id: 4, name: "Electrician", icon: "■", price: 249, time: "45 min", desc: "Switches, fans, wiring and appliance repair." },
];
export const slots = ["9:00 AM", "11:00 AM", "1:00 PM", "3:00 PM", "5:00 PM"];
```

src/App.jsx (main component)

```
import { useState } from "react";
import { services } from "../data";
import ServiceCard from "../components/ServiceCard";
import BookingForm from "../components/BookingForm";
import RegisterForm from "../components/RegisterForm";

export default function App() {
  const [page, setPage] = useState("home");
  const [selected, setSelected] = useState(null);
  const [user, setUser] = useState(null);
  const [bookings, setBookings] = useState([]);

  const addBooking = (b) => {
    setBookings([...bookings, { ...b, id: Date.now() }]);
    setSelected(null);
  };

  const cancel = (id) => setBookings(bookings.filter((b) => b.id !== id));

  return (
    <>
      <header className="nav">
        <h1 className="logo" onClick={() => setPage("home")}>Premium Fix</h1>
        <nav>
          <button className="link" onClick={() => setPage("home")}>Services</button>
          {user ? (
            <span className="hello">Hi, {user.name.split(" ")[0]}</span>
          ) : (
            <button className="btn small" onClick={() => setPage("register")}>Register</button>
          )}
        </nav>
      </header>

      {page === "register" ? (
        <RegisterForm onDone={(u) => { setUser(u); setPage("home"); }} />
      ) : (
        <main>
          <section className="hero">
            <h2>Something broke? Book a verified pro in minutes.</h2>
            <p>Plumbing, cleaning, laundry and electrical work at your door, with fixed prices.</p>
          </section>

          <section className="grid">
            {services.map((s) => (
              <ServiceCard key={s.id} service={s} onBook={() => setSelected(s)} />
            ))}
          </section>

          <section className="mybookings">
            <h3>My bookings</h3>
            {bookings.length === 0 ? (
              <p className="muted">Nothing booked yet. Pick a service above to get started.</p>
            ) : (
              bookings.map((b) => (
                <div className="row" key={b.id}>
                  <div>
                    <strong>{b.service}</strong>
                    <div className="muted">{b.date} at {b.slot} · {b.address}</div>
                  </div>
                  <button className="link danger" onClick={() => cancel(b.id)}>Cancel</button>
                </div>
              ))
            )}
          </section>
        </main>
      )}

      {selected && (
        <BookingForm
          service={selected}
          user={user}
          onClose={() => setSelected(null)}
          onRegister={() => { setSelected(null); setPage("register"); }}
        </BookingForm>
      )}
    </>
  );
}
```

```

        onConfirm={addBooking}
      />
    )}
  </>
);
}

```

src/components/ServiceCard.jsx

```

export default function ServiceCard({ service, onBook }) {
  return (
    <article className="card">
      <div className="icon">{service.icon}</div>
      <h3>{service.name}</h3>
      <p className="muted">{service.desc}</p>
      <div className="meta">
        <span>From ₹{service.price}</span>
        <span>{service.time}</span>
      </div>
      <button className="btn" onClick={onBook}>Book {service.name}</button>
    </article>
  );
}

```

src/components/BookingForm.jsx (booking form)

```

import { useState } from "react";
import { slots } from "../data";

export default function BookingForm({ service, user, onClose, onRegister, onConfirm }) {
  const [f, setF] = useState({ date: "", slot: slots[0], address: "" });
  const [err, setErr] = useState("");
  const set = (k) => (e) => setF({ ...f, [k]: e.target.value });

  const submit = (e) => {
    e.preventDefault();
    if (!f.date || f.address.trim().length < 6) {
      setErr("Choose a date and enter your full address.");
      return;
    }
    onConfirm({ ...f, service: service.name });
  };

  return (
    <div className="overlay" onClick={onClose}>
      <div className="modal" onClick={e => e.stopPropagation()}>
        <h3>Book {service.name}</h3>
        {!user ? (
          <p className="muted">Create an account first so we can confirm your booking.</p>
          <button className="btn" onClick={onRegister}>Register now</button>
        ) : (
          <form onSubmit={submit} noValidate>
            <label>Date
              <input type="date" value={f.date} min={new Date().toISOString().split("T")[0]} onChange={set("date")} />
            </label>
            <label>Time slot
              <select value={f.slot} onChange={set("slot")}>
                {slots.map((s) => <option key={s}>{s}</option>)}
              </select>
            </label>
            <label>Address
              <textarea rows="2" value={f.address} onChange={set("address")} placeholder="House no, street, area" />
            </label>
            {err && <p className="error">{err}</p>}
            <button className="btn" type="submit">Confirm booking</button>
          </form>
        )}
        <button className="link" onClick={onClose}>Close</button>
      </div>
    </div>
  );
}

```

src/components/RegisterForm.jsx (Registration Form)

```

import { useState } from "react";

const empty = { name: "", email: "", phone: "", city: "", password: "", confirm: "", terms: false };

export default function RegisterForm({ onDone }) {
  const [f, setF] = useState(empty);
  const [errors, setErrors] = useState({});

  const change = (e) => {
    const { name, value, type, checked } = e.target;

```

```

    setF({ ...f, [name]: type === "checkbox" ? checked : value });
  };

const validate = () => {
  const e = {};
  if (f.name.trim().length < 3) e.name = "Enter your full name (min 3 letters).";
  if (!/^S+@S+\S+$/i.test(f.email)) e.email = "Enter a valid email address.";
  if (!/^[6-9]\d{9}$/i.test(f.phone)) e.phone = "Enter a 10-digit mobile number.";
  if (!f.city) e.city = "Select your city.";
  if (f.password.length < 6) e.password = "Password must be at least 6 characters.";
  if (f.confirm !== f.password) e.confirm = "Passwords do not match.";
  if (!f.terms) e.terms = "Accept the terms to continue.";
  return e;
};

const submit = (ev) => {
  ev.preventDefault();
  const e = validate();
  setErrors(e);
  if (Object.keys(e).length === 0) onDone(f);
};

const Field = ({ name, label, type = "text", ...rest }) => (
  <label>{label}
    <input name={name} type={type} value={f[name]} onChange={change} {...rest} />
    {errors[name] && <span className="error">{errors[name]}</span>}
  </label>
);

return (
  <main className="auth">
    <form className="panel" onSubmit={submit} noValidate>
      <h2>Create your account</h2>
      <p className="muted">One account to book, track and cancel services.</p>
      <Field name="name" label="Full name" />
      <Field name="email" label="Email" type="email" />
      <Field name="phone" label="Mobile number" type="tel" maxLength={10} />
      <label>City
        <select name="city" value={f.city} onChange={change}>
          <option value="">Select city</option>
          {[ "Hyderabad", "Bengaluru", "Chennai", "Mumbai", "Delhi" ].map((c) => <option key={c}>{c}</option>)}
        </select>
        {errors.city && <span className="error">{errors.city}</span>}
      </label>
      <Field name="password" label="Password" type="password" />
      <Field name="confirm" label="Confirm password" type="password" />
      <label className="check">
        <input type="checkbox" name="terms" checked={f.terms} onChange={change} />
        I agree to the terms and privacy policy
      </label>
      {errors.terms && <span className="error">{errors.terms}</span>}
      <button className="btn" type="submit">Create account</button>
    </form>
  </main>
);
}

```

src/styles.css

```

:root { --ink:#0F2A43; --bg:#F2F6F9; --card:#fff; --gold:#F2B134; --teal:#1F8A70; --muted:#5B6B7B; --red:#C0392B; }
* { box-sizing: border-box; }
body { margin:0; font-family:"Manrope",sans-serif; background:var(--bg); color:var(--ink); }
h1,h2,h3 { font-family:"Bricolage Grotesque",sans-serif; margin:0; }
.nav { display:flex; justify-content:space-between; align-items:center; padding:16px 6vw; background:var(--ink);
  color:#fff; position:sticky; top:0; z-index:5; }
.nav nav { display:flex; align-items:center; gap:16px; }
.logo { font-size:1.5rem; cursor:pointer; }
.logo::after { content:""; display:inline-block; width:10px; height:10px; background:var(--gold); border-radius:50%;
  margin-left:6px; }
.hello { color:var(--gold); font-weight:700; }
main { padding:0 6vw 60px; }
.hero { padding:56px 0 32px; max-width:640px; }
.hero h2 { font-size:clamp(1.9rem,5vw,3rem); line-height:1.1; }
.hero p { color:var(--muted); font-size:1.05rem; }
.grid { display:grid; grid-template-columns:repeat(auto-fit,minmax(230px,1fr)); gap:20px; }
.card { background:var(--card); border-radius:16px; padding:22px; display:flex; flex-direction:column; gap:10px;
  border:1px solid #dde6ee; }
.icon { font-size:2rem; width:56px; height:56px; display:grid; place-items:center; background:#FFF4D9; border-
  radius:14px; }
.meta { display:flex; justify-content:space-between; font-weight:700; color:var(--teal); margin-top:auto; }
.muted { color:var(--muted); margin:0; font-size:.92rem; }
.btn { background:var(--card); color:var(--ink); border:0; padding:12px 18px; border-radius:10px; font:700 .95rem
  "Manrope"; cursor:pointer; }
.btn:hover { filter:brightness(.95); }
.btn.small { padding:8px 14px; }
.link { background:none; border:0; color:inherit; font:600 .95rem "Manrope"; cursor:pointer; }
.link.danger { color:var(--red); }

```

```

:focus-visible { outline:3px solid var(--teal); outline-offset:2px; }
.mybookings { margin-top:44px; }
.row { display:flex; justify-content:space-between; align-items:center; background:var(--card); padding:14px 18px;
  border-radius:12px; margin-top:10px; border-left:5px solid var(--teal); }
.overlay { position:fixed; inset:0; background:rgba(15,42,67,.6); display:grid; place-items:center; padding:16px;
  z-index:10; }
.modal, .panel { background:#fff; border-radius:18px; padding:26px; width:100%; max-width:440px; display:flex; flex-
  direction:column; gap:14px; }
.modal form { display:flex; flex-direction:column; gap:14px; }
.auth { display:grid; place-items:center; padding-top:40px; }
label { display:flex; flex-direction:column; gap:6px; font-weight:600; font-size:.9rem; }
input, select, textarea { padding:11px 12px; border:1.5px solid #cbd7e2; border-radius:10px; font:inherit;
  background:#fff; }
input:focus, select:focus, textarea:focus { border-color:var(--teal); outline:none; }
.check { flex-direction:row; align-items:center; gap:10px; font-weight:400; }
.check input { width:18px; height:18px; }
.error { color:var(--red); font-size:.82rem; font-weight:600; margin:0; }

```

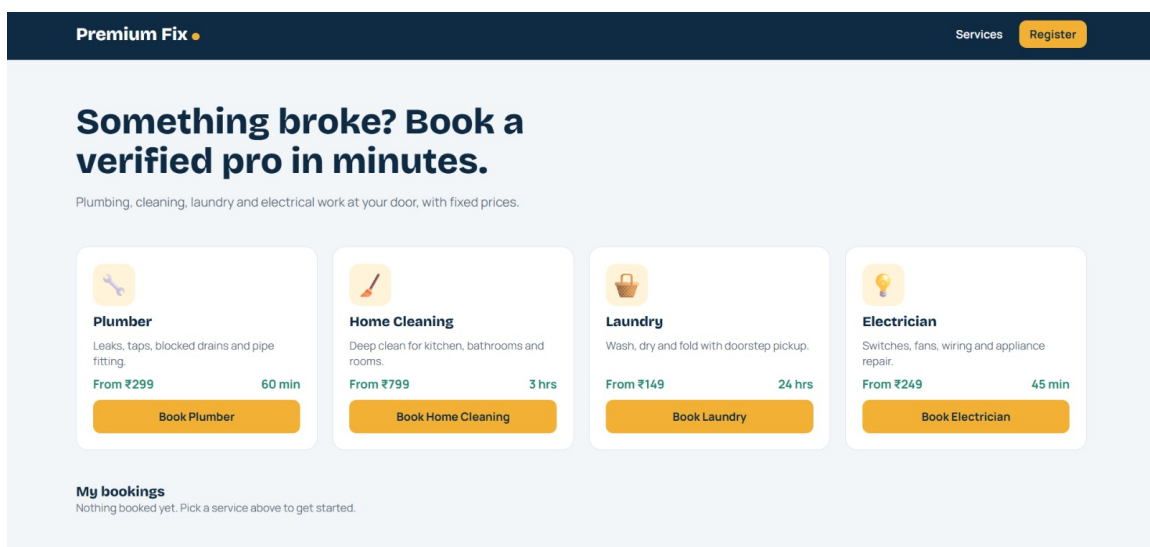
Details about the Program

Where Saved	All files are saved inside the project folder FL_LAB_ANITS/fs_week_9/ in the GitHub repository raviteja-codehub/FL_LAB_ANITS — package.json, vite.config.js, index.html and the src/ folder with main.jsx, App.jsx, data.js, styles.css and the components/ folder (ServiceCard.jsx, BookingForm.jsx, RegisterForm.jsx). node_modules is not uploaded.
Where Executed	The application runs on the Vite development server (Node.js) from the terminal on the local computer and is opened in a web browser (Google Chrome / Edge / Firefox) at http://localhost:5173.
How Executed	1. Open a terminal in the fs_week_9 folder. 2. Run npm install to install React and Vite. 3. Start the app with npm run dev; the terminal prints the local URL http://localhost:5173/. 4. Open the URL in the browser, click Register, fill the form and create the account, choose a service, book it and view it under My bookings. Use Cancel to remove a booking.

Output

After running npm run dev and opening http://localhost:5173, the landing page, registration form and booking pages appear as shown below:

Landing Page (services list)



Registration Page (Registration Form)

Premium Fix

Services Register

Create your account

One account to book, track and cancel services.

Full name

Raviteja Pasupuleti

Email

ravi@gmail.com

Mobile number

886698943

City

Hyderabad

Password

Confirm password

☒ I agree to the terms and privacy policy

Create account

Service Booking Form (Electrician)

Premium Fix

Services Hi, Raviteja

Something broke? Book a verified pro in minutes.

Plumbing, cleaning, laundry and electrical work at your door, with fixed prices.

Plumber

Leaks, taps, blocked drains and pipe fitting.

From ₹299 60 min

Book Plumber

Home cleaning

Deep cleaning, carpet cleaning, etc.

From ₹149 1 hr

Book Home cleaning

Electrician

Switches, fans, wiring and appliance repair.

From ₹249 45 min

Book Electrician

My bookings

Nothing booked yet. Pick a service above to get started.

Book Electrician

Date

10/25/2026

Time slot

9:00 AM

Address

2/43, ANITS Road, Vizag

Confirm booking

Close

My Bookings (after confirming the booking)

Premium Fix

ServicesHi, Raviteja

Something broke? Book a verified pro in minutes.

Plumbing, cleaning, laundry and electrical work at your door, with fixed prices.



Plumber
Leaks, taps, blocked drains and pipe fitting.
From ₹29960 min
[Book Plumber](#)



Home Cleaning
Deep clean for kitchen, bathrooms and rooms.
From ₹7993 hrs
[Book Home Cleaning](#)



Laundry
Wash, dry and fold with doorstep pickup.
From ₹14924 hrs
[Book Laundry](#)



Electrician
Switches, fans, wiring and appliance repair.
From ₹24945 min
[Book Electrician](#)

My bookings

Electrician

2026-10-25 at 9:00 AM - 2/43, ANITS Road, Vizag

Cancel